

Maciej Tkacz

Robotics Engineer - Industrial Robot Programming, Simulation & Digital Twins

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PROFESSIONAL PROFILE

Robotics engineer combining industrial robot programming with simulation and hands-on machine building. Currently a Robotics Engineer at ASTOR, where I program Kawasaki and Epson robots, build digital twins of production lines and robot simulations, and design and build demonstration stations presenting those robots across different applications. Previously an Application Engineer at AIAutomation, creating robotics simulations and digital twins of production workcells in Visual Components for automotive clients, including Python scripts for process optimization. Mechanical Engineering student at Cracow University of Technology and certified Software Technician, so mechanics, electronics and code come from the same background. Designed and built a SCARA robotic arm end-to-end - CAD, 3D printing, stepper drives, microcontroller electronics and Python/C++ control software with forward and inverse kinematics. Based in Krakow, available full-time, Polish native, English C1.

KEY SKILLS

- Industrial robot programming: Kawasaki and Epson robots - teach pendant workflows, motion programs, complete application cycles, cell testing and commissioning; certified integrator course (ASTOR Robotics Center) and Epson programming course.
- Robotics simulation & digital twins: digital twins of production lines and workcells, robot paths and motion sequences, process logic, collision checking, cycle flows, layout optimization and virtual validation before physical implementation; Visual Components used commercially for automotive clients.
- Programming: Python (automation, process optimization, simulation scripting, calculations, data processing), C++ and Arduino/C for hardware control, JavaScript basics; iterative debugging and technical troubleshooting.
- Robot cells & demonstration systems: design, assembly, integration, programming and commissioning of robot stations presenting industrial robots in different applications, across mechanics, electrics and robot software.
- Mechanical design & CAD: SolidWorks, Autodesk Inventor, Fusion 360, AutoCAD; design for manufacturing and additive manufacturing, reverse engineering, geometry preparation for simulation.
- Prototyping & electromechanics: FDM, SLA and MJF 3D printing including PET-G and carbon-fibre reinforced materials; NEMA stepper motors, TMC/DRV drivers, Arduino/ESP32-class microcontrollers, wiring and bring-up testing.
- Engineering fundamentals: kinematics, control systems, dynamic systems modelling, mechatronics and analytical mechanics from university; MATLAB from coursework and ROS from personal robotics projects.
- Working style: technical documentation, communication with engineers, customers and stakeholders, and independent problem solving in interdisciplinary teams; Polish native, English C1, driving licence category B.

SELECTED ROBOTICS PROJECT

SCARA Robotic Arm Prototype | Personal R&D Project

- Designed and built an end-to-end SCARA robotic arm prototype, covering mechanical concept, CAD modelling, kinematic layout, selection of NEMA stepper actuators and TMC/DRV drivers, microcontroller electronics and wiring.
- Optimized components for 3D printing, including PET-G and carbon-fibre reinforced materials, with focus on stiffness, assembly constraints and rapid iteration.
- Developed Python/C++ control software for motion and hardware-interaction experiments, applying forward/inverse kinematics and control-system fundamentals.
- Integrated mechanical, electronic and software subsystems through iterative assembly, testing and debugging, gaining practical hardware-software integration experience.

PROFESSIONAL EXPERIENCE

Robotics Engineer | ASTOR, Krakow | Jul 2026 - present

- Program Kawasaki and Epson industrial robots, from motion programs and teach pendant work to complete application cycles.
- Build digital twins of production lines and robot simulations, validating reach, motion sequences and process behaviour before physical implementation.
- Design, prepare and build demonstration stations presenting Kawasaki and Epson robots across different applications - mechanical assembly, hardware-software integration, programming and commissioning.
- Test and troubleshoot robot cells end to end, working across mechanics, electrics and robot software.

Application Engineer | AIAutomation | Jan 2025 - May 2026

- Created robotics simulations and digital twins of production workcells in Visual Components for automotive clients.
- Developed Python scripts for process optimization and for improving simulation behaviour and performance.
- Developed robot logic, motion sequences, robot paths, collision checks, cycle flows and virtual process validation.
- Optimized workcell layouts and 3D geometry for simulation needs, using SolidWorks and Autodesk Inventor.
- Analyzed customer documentation, engineering standards and technical specifications to support solution concepts aligned with production requirements.
- Collaborated with engineers and stakeholders, communicating simulation assumptions, constraints and improvement proposals.

3D Printing and CAD Design Specialist | Cubic Inch Additive Manufacturing, Piaseczno | Jun 2023 - Aug 2023

- Operated and serviced FDM, MJF and SLA 3D printers, supervising process parameters, post-processing and quality control.
- Designed and optimized CAD models in Fusion 360 and Autodesk Inventor for additive manufacturing and rapid prototyping.

- Supported implementation testing for a new SLA technology, documenting progress and technical observations.
- Coordinated production tasks in a 10-person project team, balancing quality, manufacturability and delivery constraints.

Robotics and 3D Printing Intern | ASTOR Robotics Center, Krakow | May 2022

- Assembled mechanical equipment and 3D-printed components for Kawasaki robots and Astorino educational robot platforms.
- Programmed Kawasaki industrial robots using teach pendant workflows and created basic robot motion programs.
- Tested robot and workstation operation, gaining practical exposure to robot setup, safety and hardware integration.

Web Application Development Intern | Souczek Design Studio Reklamy i Druku, Kielce | Jul 2021

- Solved technical problems in JavaScript, including implementation of an interactive order form.
- Tested and deployed web functionality in a team environment according to customer guidelines.

EDUCATION

- Cracow University of Technology - Mechanical Engineering, Engineer's degree in progress (Oct 2024 - present).
Relevant coursework: dynamic systems modelling, automation/control, analytical mechanics and mechatronics.
- PKMechPower Student Research Group, Mechanical Section - CAD and 3D printing tasks, including driver seat and brake-system components; mass and strength-oriented design optimization.
- Zespol Szkol Informatycznych im. gen. Jozefa Hauke-Bosaka, Kielce - Software Technician (Sep 2019 - Apr 2024).

CERTIFICATES

- Kawasaki robot operation and programming - integrator course with certificate, ASTOR Robotics Center (May 2022).
- Epson industrial robot programming - certified course.
- Python programming courses and self-directed Python/C++ development in robotics and automation projects.

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